Recap: Game Theory
Finding Nash Equilibria
Classic Games

A "troubling" example: matching pennies

② → mismatch

	heads	tails
heads	1, -1	-1, 1
tails	-1, 1	1, -1

① → match

these games have no Nash Equilibrium in pure strategies
↳ a specific choice of an action

these games do have equilibria in mixed strategies

mixed strategies: a choice among actions according to some probabilities

here: heads with probability $\frac{1}{2}$
tails with probability $\frac{1}{2}$

why? if we are player 1 and player 2 plays this way:

expected payoff heads: $(\frac{1}{2})(1)+(\frac{1}{2})(-1)=0$
expected payoff tails: $(\frac{1}{2})(-1)+(\frac{1}{2})(1)=0$

① is different between their two strategies so playing a mixed strategy makes sense

Finding a Mixed Strategy Nash Equilibrium

offense

defense	run $^{(p)}$	pass $^{(1-p)}$
q blitz	2, 0	0, 4
1-q zone	0, 1	1, 0

again, no pure strategy Nash Equilibrium in this game

To find mixed strategy Nash Equilibrium (NE)

→ use the fact that in mixed strategy NE, players are indifferent between their pure strategies

step 1) Find probalities for offense that make defense indifferent between blitz & zone

→ probability of run = p, probability of pass = $1-p$

defense: expected payoff from blitz = $(2)(p)+(0)(1-p) = 2p$

expected payoff from zone = $(0)(p)+(1)(1-p) = 1-p$

indifference: $2p = 1-p$

$p = \frac{1}{3}$

so in mixed strategy NE, offense plays run with probability $\frac{1}{3}$, pass with probability $\frac{2}{3}$

step 2) Find probabilities for defense that makes offense indifferent between run and pass

→ probality of blitz = q, probability of zone = $1-q$

offense: expected payoff from run = $(0)(q)+(1)(1-q) = 1-q$

expected payoff from pass = $(4)(q)+(0)(1-q) = 4q$

indifference: $1-q = 4q$

$q = \frac{1}{5}$

so in mixed strategy NE, defense plays blitz with probability $\frac{1}{5}$, zone with probability $\frac{4}{5}$

This is the only Nash Equilibrium of this game

Repeated Games

Remember the Prisoners' Dilemma

		② cheat	② coop
①	cheat	1, 1	4, 0
	coop	0, 4	3, 3

→ unique Nash Equilibrium is (cheat, cheat)

what if these players repeated this game over and over? could they cooperate? → maybe

<u>new feature</u>: discounting

to make this more realistic, we assume that players "discount" future payoffs
→ care more about payoffs right now

	now	next round	two rounds from now	three rounds from now
payoff of 1 is worth:	1	$1 \cdot r$	$1 \cdot r^2$	$1 \cdot r^3$

discount r discount r discount r

r is some number between zero and one

if r is close to 1: players are very patient

if r is close to 0: players are very impatient